

BIO 44E3. 17 NOV 2025

parasites manipulate sex ratios
VERTICALLY transmitted (maternal).

- male-killing.
- feminization.

Gammarus . infected with Octosporea effeminans

90% of offspring of parasitized moms are ♀.

50-80% in unparasitized broods.

Wohlbachia - intracellular bacteria.

(insect parasites)

pill woodlouse .

WZ = females

ZZ = males.

Wohlbachia is feminizing.

the W chromosome has been

lost from some populations.

everybody is ZZ,

Wohlbachia-inf are female.

- Sexual selection .

secondary sex characteristics

peacock's tails, bird song, color, displays . . .

- male-male competition .
- + female choice .

FISHER - runaway selection
'sexy son'

- preference + trait become genetically correlated .
- trait is expressed in males
- preference is " " females

HERITABLE trait . initial preference .

trait can't be too expensive .

- PREDICTIONS:

- zero or negative corr between male quality + trait
- genetic correlation between trait + preference
- high variability in traits are preferred

- HONEST SIGNALING .

male quality

for GOOD GENES model .

how to prevent cheating?

- COSTLY traits?

- REVEALING trait .

- CONDITIONAL trait

- handicap traits - more expensive or less beneficial for low-quality males.

- SIGNALING .

- HAMILTON - ZUK hypothesis

- ♀ are selecting ♂ with parasite resistance genes?

- females prefer showy males

- correlation between trait + ♂ quality .

- heritable paras resistance

- parasite load significantly decrease host viability

positive correlation between parasite load & showiness of a species -
ectoparasite load vs human judgement of showiness

- reverse causality ?

- other factors .

- phylogenetic inertia .

WITHIN-SPECIES .

- guppies .

- parasite infestation lowers brightness
- reduce number of sexual displays
- heritable?
- can't rule out transmission avoidance

BARN SWALLOWS .

♀ prefer ♂ with longer tails .

long-tailed ♂ have fewer mites

chicks with more mites are smaller at fledging

- CROSS-FOSTERING .

offspring of long-tailed ♂ have fewer mites .

rules out luck, direct benefits .

direct benefits
selection for parental care
selection for parasite avoidance